

Spatial Statistics in Epidemiology and Public Health

Lecture 9: Disease Ecology

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Disease Ecology: What do we want to do?

Pattern and Process

Point Processes: Chagas Disease in Peru

Cluster detection

Spatial relative risk

Spatial Regression: Raccoon Rabies

Statistical estimation of landscape barriers

Wombling

Spatially varying coefficients

Modeling surveillance

Disease dynamics

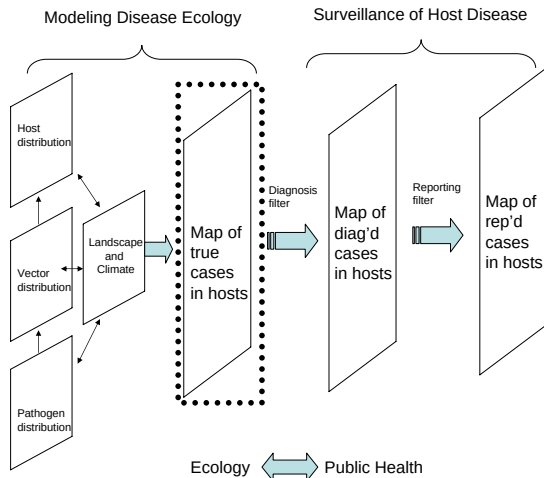
Disease Ecology

- ▶ Interactions between virus, host, landscape.
- ▶ Landscape epidemiology (Pavlovsky, 1967), landscape ecology (Manel et al. 2003, *TrEE*), spatial epidemiology (Osfeld et al. 2005, *TrEE*), landscape genetics (host and virus) (Biek et al. 2006, *Science*), conservation medicine (Aguirre et al. 2002).
- ▶ People, animals, diseases, ecology, environment!
- ▶ Spatio-temporal data, mathematical models, genetic sequences, missing data, GIS!

Epizootology and Epidemiology

- ▶ Most emergent infectious diseases have animal reservoir (WNV, Ebola, Avian influenza, Monkeypox, SARS, HIV/SIV).
- ▶ History of animal/human disease (Torrey and Yolken, 2005, *Beasts of the Earth*).
- ▶ Interesting intersection of modelers, ecologists, statisticians, medical geographers, ecological geneticists, public health researchers, epidemiologists.
- ▶ One Health.

The “big picture”



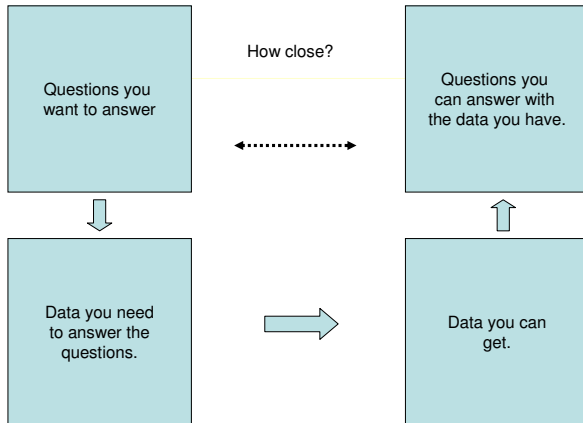
Pattern and Process

- ▶ Our ultimate goal is understanding the ecological processes driving the patterns we see in our observations.
- ▶ When linking process (model or reality) to pattern (data), typically:
 - ▶ Ecology focus: Process to pattern
 - ▶ Emphasis on mathematical model, link to available data
 - ▶ Statistics: Pattern to process
 - ▶ Collected data, hypothesis test or analytic (e.g., regression) model.

Ultimately futile exercise?

- ▶ Process may not yield unique pattern (e.g., chaos, stochasticity).
- ▶ Pattern may not reveal unique process without additional information (e.g., spatial point patterns, Bartlett (1964)).
- ▶ But the real question is, “Can we learn more than we already know?”
- ▶ If not, what additional data do we need?

The whirling vortex



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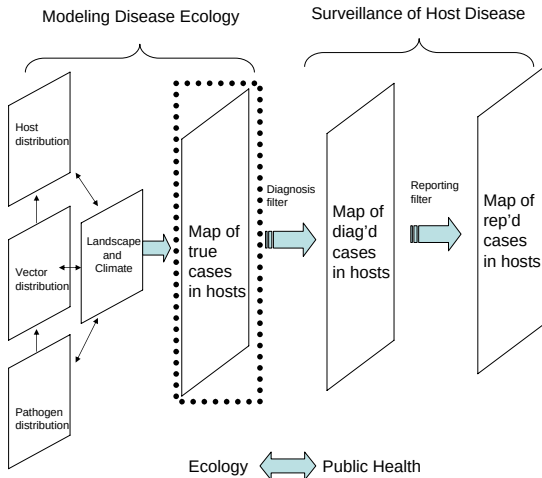
Point Processes in Disease Ecology: Chagas disease in Peru

- ▶ Joint work with Michael Levy (University of Pennsylvania)
- ▶ Chagas disease: Vector borne disease (infection with *T. cruzi*).
- ▶ Vector (in southern Peru): *Triatoma infestans*.
- ▶ Study area: Guadalupe, Peru (peri-urban).
- ▶ Fields surrounding rocky hilltops with houses.
- ▶ GPS all household locations.
- ▶ Spraying campaign, identify house locations, houses with vectors (“infested”), and houses with infected vectors (“infected”).

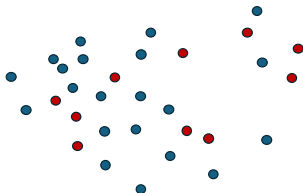
How to find a cluster?

- ▶ Consider two approaches: scan statistic and intensity estimators.
- ▶ Spatial scan statistic:
 - ▶ Define set of potential clusters (elements of scanning window).
 - ▶ Assign “score” to each potential cluster.
 - ▶ Find “most likely cluster” (MLC) as potential cluster with extreme score.
 - ▶ Evaluate significance of most likely cluster via Monte Carlo simulation.
 - ▶ Compare observed “score” of MLC to distribution of scores MLCs (regardless of location) under random assignment.
 - ▶ SaTScan software (www.satscan.org).

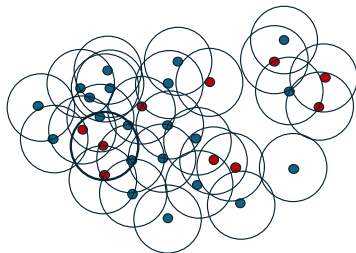
Where are we in the big picture?



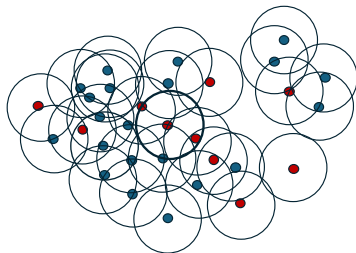
SaTScan example: Observed data



SaTScan example: Most likely cluster

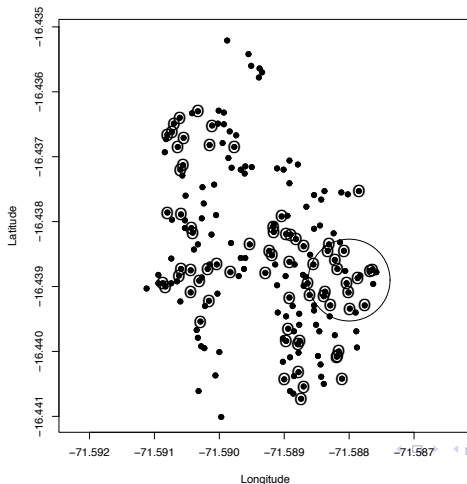


Simulation 1 (locations fixed, 10 red, location random)



SaTScan, Infected among infested, Most likely cluster ($p=0.181$)

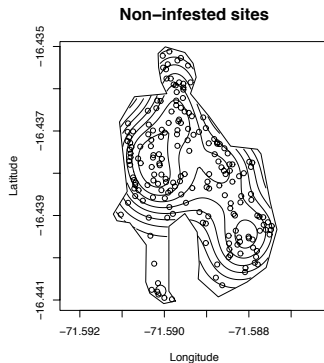
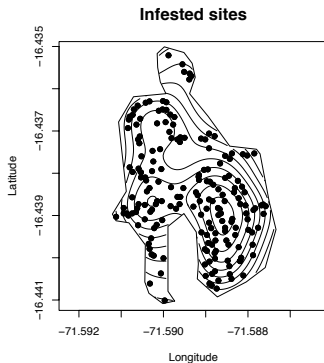
SaTScan, Most Likely Cluster, Infected, p -value = 0.181



Chagas SaTScan conclusions

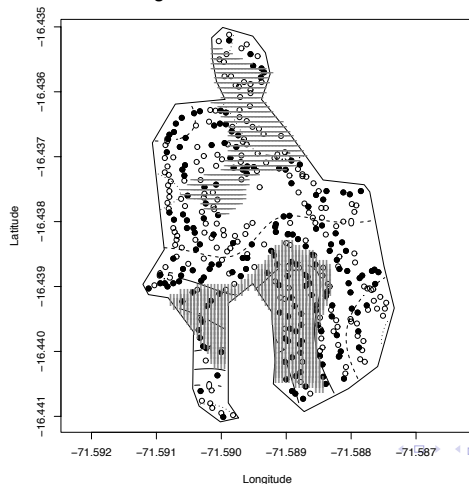
- ▶ Statistically significant cluster of infested households among all households.
- ▶ No statistically significant cluster of infected households among infested households.
- ▶ Note circular most likely cluster may include gaps (top of hill).
- ▶ What about non-circular clusters?

Kernel intensity estimates, infested vs. all households

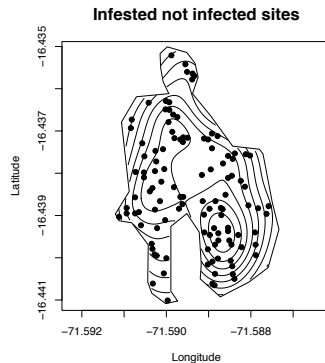
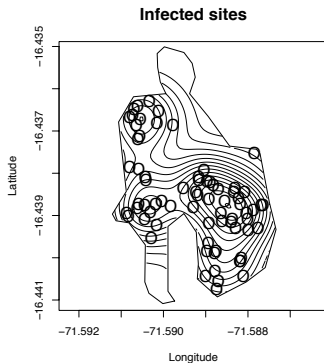


Ratio of kernel intensity estimates, infested vs. all households

Log relative risk surface: Infested

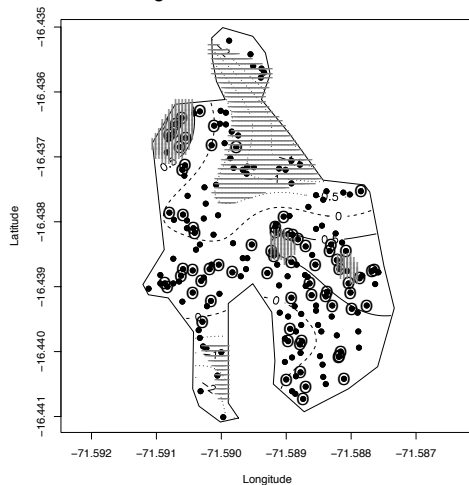


Kernel intensity estimates, infected vs. infested households



Ratio of kernel intensity estimates, infected vs. infested

Log relative risk surface: Infected



Cluster conclusions

- ▶ Relative risk surface adds more geographical precision to patterns initially revealed by SaTScan.
- ▶ Large risk of infestation in the south.
- ▶ Within this some pockets of increased risk of infection.
- ▶ Area of lower risk missed by circular scan statistic, due to its irregular shape.
- ▶ Identifies areas for future studies.

Chagas conclusions

- ▶ Significant cluster of infested households, but no clusters of infected households (circular clusters).
- ▶ Relative risk surface also suggests area of low risk (both infestation *and* infection) in northeast.
- ▶ K functions suggest significant *clustering* of *infected* but not *infested* households.
- ▶ Taken together, results reveal different aspects of the underlying process.
- ▶ A single cluster does not define clustering, nor does clustering imply a single cluster.

Chagas conclusions

- ▶ Infestation: pockets of higher and lower relative risk, but level of clustering not different between cases and controls.
- ▶ Infection: More clustered at small distances than infestation, but resulting clusters are smaller and more diffuse.
- ▶ Scale of clustering different between infestation and infection, and larger than typical range of individual vectors.
- ▶ Scale of clustering useful in targeted surveillance for human cases (Levy et al., 2007, *PLoS NTD*).

Questions?

Raccoon rabies



What is rabies?

- ▶ Virus in family of Lyssa (“frenzy”) virus.
- ▶ Reportable disease.
- ▶ Various strains associated with primary host (bat, dog, coyote, fox, skunk, and raccoon).
- ▶ Host cross-over, typically transmitted via bite/scratch.
- ▶ Most human infection from bat strains.

Raccoon rabies

- ▶ Endemic in Florida and South Georgia.
- ▶ Translocation of rabid animal(s) to VA/WV border circa 1977.
- ▶ Wave-like spread since.
- ▶ Connecticut first appearance 1991-1996.
- ▶ Ohio 2005.
- ▶ Joint work with Leslie Real's lab in Population Biology, Evolution, and Ecology (David Smith, Colin Russell, Roman Biek, Scott Duke-Sylvester).

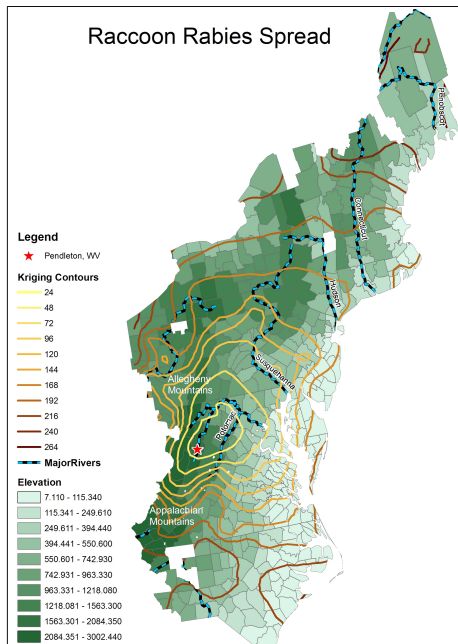
Barrier estimation: What do we want?

- ▶ Goal: Measure effect of landscape features, (e.g., mountains and rivers) on the speed of raccoon rabies diffusion.
- ▶ Elevation, river or road presence significantly related to raccoon rabies counts (Recuenco et al. 2007) and transmission time (Russell et al. 2004).
- ▶ Landscape features may serve as either barriers or gateways to the spread of infectious disease.
- ▶ Find and visualize barriers: Do they align with certain landscape features?

Data: What do we have?

- ▶ Time in months to first reported raccoon rabies case in 428 contiguous counties in the Eastern US (CDC).
 - ▶ 0 for origin county: Pendleton, WV.
- ▶ Mean elevation by county (USGS - Geographic Names Information System).
- ▶ Indicator for major river presence in county (ESRI data and a geographic information system (GIS)).
- ▶ Population density by county (US Census and ESRI).
- ▶ Distance between origin county and all counties.

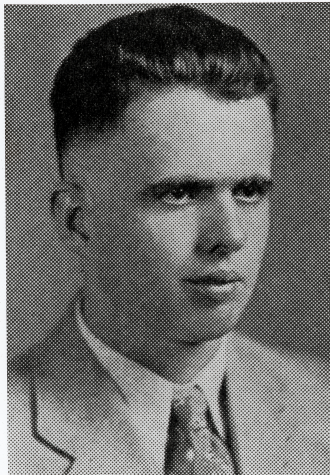
Data



Wombling

- ▶ Joint work with David Wheeler (Wheeler and Waller, *JABES*, 2008).
- ▶ Wombling: determine boundaries on a map by finding where local spread (change) is slower than elsewhere (Womble, 1951 *Science*).
- ▶ William H. Womble a bit of an elusive figure...

William H. Womble (?)



Google search: W.H. Womble Professor Robert Stencel



Which leads to...



Are you weady to womble?

- ▶ Consider a set of potential boundaries and decide if each is a “real” boundary or not.
- ▶ Many algorithmic approaches both deterministic and “fuzzy”.
- ▶ Adopt a Bayesian hierarchical model for wombling (Lu and Carlin 2005).
- ▶ Bayesian approach provides a direct estimate of the probability that a line segment between two adjacent areas is a barrier (fuzzy boundary) in contrast to algorithmic versions based on thresholds, etc.

Bayesian areal wombling

- Model time to first reported raccoon rabies case Y_i :

$$Y_i | \mu_i, \tau \sim N(\mu_i, 1/\tau)$$

where

$$\mu_i = \alpha + \phi_i$$

is the expected value of time to first case per county.

- Spatial random effects follow a conditionally autoregressive (CAR) prior $\phi \sim CAR(\eta)$ with a mean random effect determined by its neighboring values.

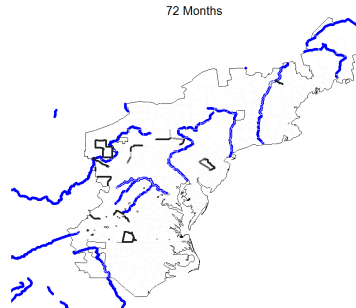
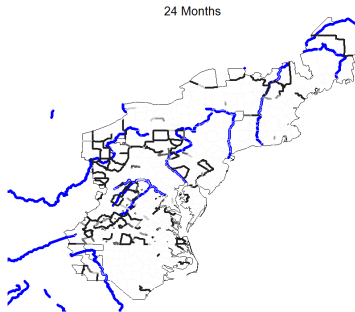
Bayesian areal wombling

- ▶ *Boundary likelihood value* (BLV) assigned to each potential boundary (here, edge between two counties), based on difference in expected (modeled) time to first appearance.

$$\Delta_{ij} = |\mu_i - \mu_j|$$

- ▶ Use MCMC to draw sample from posterior $[\Delta_{ij}|\mathbf{y}]$ based on draws from posteriors $[\mu_i|\mathbf{y}]$ and $[\mu_j|\mathbf{y}]$.
- ▶ This assigns a posterior probability for each edge, then display edges with with $p(\Delta_{ij} > c|\mathbf{y})$ for some threshold probability c .

Wombling boundaries: $p(\Delta_{ij} > c | y)$



Linking to local covariates

- ▶ Bayesian areal wombling provides estimates of barriers but does not allow direct inference regarding the impact of particular landscape barriers on the evidence for barriers.
- ▶ We could expand our fixed effect α to $\mathbf{X}'\beta$ to include local covariates (e.g., elevation, boundary based on a river).
- ▶ However, what if the effect of elevation or presence of river varies from place to place?
- ▶ Russell et al. (2003, *PNAS*) suggest that river effect depends on direction of movement of the wave (perpendicular? Slower. Parallel? Faster.)

Spatially varying coefficients

- ▶ We consider a *spatially varying coefficient* model with CAR priors on the covariate effects β , i.e.,

$$Y_i | \mu_i, \tau \sim N(\mu_i, 1/\tau)$$

where

$$\mu_i = \mathbf{X}_i' \beta_i + \phi_i$$

- ▶ Spatial priors on elements of β_i .
- ▶ More specifically, assign a multivariate CAR prior on the set of β (Banerjee et al. 2004).

MultiCAR details

- ▶ $\beta_i = (\beta_{i1}, \beta_{i2}, \dots, \beta_{ip})'$
- ▶ $\beta_i | (\beta_{(-i),1}, \beta_{(-i),2}, \dots, \beta_{(-i),p}) \sim N(\bar{\beta}_i, \Omega/m_i)$
where

$$\bar{\beta}_i = (\bar{\beta}_{i1}, \bar{\beta}_{i2}, \dots, \bar{\beta}_{ip})'$$

and

$$\bar{\beta}_{i1} = \sum_{k \in \kappa_i} \beta_{k1} / m_i$$

where κ_i = neighbor set for region i , and $|\kappa_i| = m_i$.

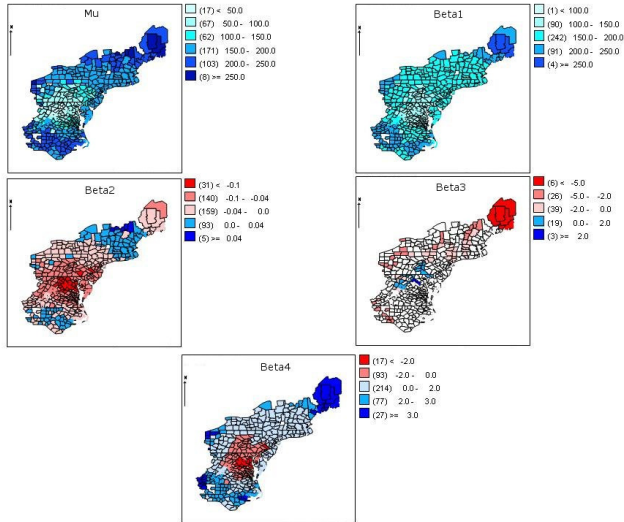
- ▶ $\Omega \sim \text{Inverse-Wishart}(\nu, 0.02 \cdot I_{p \times p})$.

Including covariates

- ▶ Include effects of (mean) elevation, presence of a major river, and the natural log of the (human) population density.
- ▶ Best fitting (via DIC) model includes spatial variation in all three (and intercept).

$$E[Y_i] = \beta_{i1} + \beta_{i2}(\text{mean elev}) + \beta_{i3}(\text{river}) + \beta_{4i}(\log(\text{pop dens}))$$

β_1 : int, β_2 : elev, β_3 : river, β_4 : log pop



Findings/interpretations

- ▶ Map of posterior mean (MU): shows the overall wave or spread.
- ▶ Random intercept reveals local adjustments.
- ▶ River effect indicates increases in time until first appearance across Potomac and Susquehanna Rivers, decreases time for Hudson River and others.
- ▶ Elevation is not difference in elevation so not directly informing on elevation gradients as barriers, simply elevation impact on time until appearance.

SVC wombled boundaries: $p(\Delta_{ij} > c | \mathbf{y})$

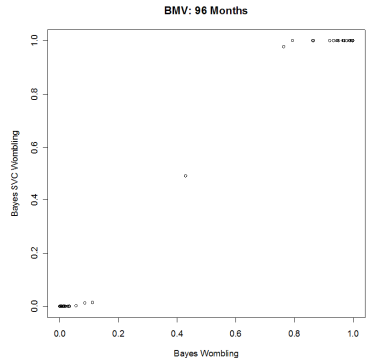
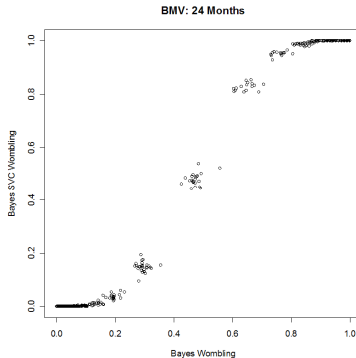
24 Months



72 Months



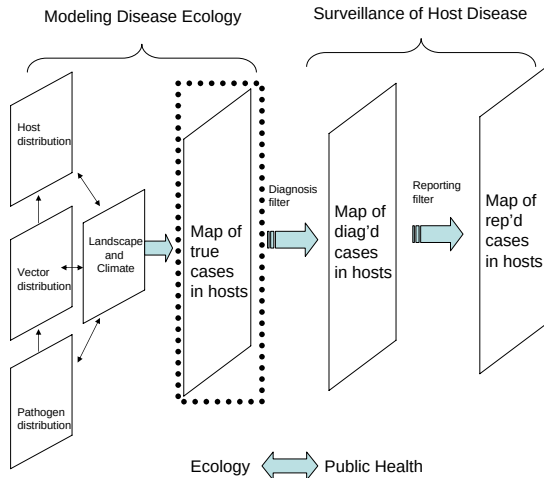
Including covariates → better wombling?



Next steps: Surveillance

- ▶ WHO: Surveillance is “on ongoing, systematic collection, analysis, and interpretation of health-related data essential to planning, implementation, and evaluation of public health practice”.
- ▶ How do we detect an outbreak as it is happening?
- ▶ What data do we have?
- ▶ Can the data tell us where to target increased surveillance efforts as well as what is going on?
- ▶ Back to the “big picture”.

Where are we in the big picture?



What are our questions?

- ▶ What is an epidemic?
- ▶ Above a baseline?
- ▶ Here: any occurrence of an infectious disease detected in a novel geographic location that poses a public health risk.
- ▶ Want to spot new cases in new places to plan prevention and response.
- ▶ Challenge: Surveillance of animal reservoirs.

Reality check

- ▶ Ill raccoon in my back yard last year. Dead in morning.
Thought: Animal control might want to know and test for rabies.
- ▶ Algorithm:
 - ▶ Call animal control: “Unless it bit you or your pets, we don’t respond to dead animals.”
 - ▶ Call sanitation: “We won’t come into your yard but we can schedule curbside pick-up.”
 - ▶ Call poison control (as suggested on CDC website): “Sounds rabid, don’t touch it. Call animal control, they will want to know.”
 - ▶ Repeat.
- ▶ Result: No testing, no data.

Assessing surveillance

- ▶ Can we use disease dynamics and observed data to assess:
 - ▶ Reporting rate (probability of reporting)?
 - ▶ Reporting efficiency (proportion of cases reported)?
- ▶ These are not exclusive concepts and they might vary in space and time.

Gerardo-Giorda et al. 2013, *J R Soc Interface*

- ▶ Raccoon rabies in New York State.
- ▶ SIR (actually SEI) model + model of surveillance (function of reported cases).
- ▶ Goal: How to use reporting data (positive and negative occurrences) to identify geographic areas where surveillance levels are potentially insufficient to detect outbreaks.
- ▶ Two approaches: constant reporting rate and time-varying reporting rate.

Assumptions

- ▶ Susceptibles (S), Exposed (E), and Infectious (I) model.
- ▶ σE = rate of new infections.
- ▶ Individual in I class for a very short time ($1/\sigma$).
- ▶ No reproduction in I class, but reproduction in S and E .
- ▶ Density dependent mortality for S and E (not I)

Dynamics

- S (healthy) E (latent) I (infectious) model:

$$S' = a(S + E) - bNS - \beta IS,$$

$$E' = \beta IS - bNE - \sigma E,$$

$$I' = \sigma E - \alpha I,$$

and

$$N = S + E + I.$$

That is:

$$S' = (\text{births from } S \text{ and } E) - (\text{deaths of } S) - (S \rightarrow E),$$

$$E' = (S \rightarrow E) - (\text{deaths of } E) - (E \rightarrow I),$$

$$I' = (E \rightarrow I) - (\text{deaths of } I).$$

Aggregate (reduce) to model of N and I

- ▶ Little information on E state (not observed).
- ▶ Aggregate to model of N and I (maintains essential dynamics, assessed via simulation).

$$\begin{aligned}N' &= aN - (a + \alpha)I - bN(N - I) \\I' &= -\alpha I + \sigma E\end{aligned}$$

- ▶ We don't observe σ or E , but we can (partially) observe the number of new infectious animals.
- ▶ Replace σE by Φ defining number of new members of I .
- ▶ Estimate Φ by $F(R_+, R_-)$, function of reported positive and negative cases.

What do we know about raccoon rabies dynamics?

- ▶ Birth rate, contact rate, latency, infectious period, death rate.
- ▶ $R_0 \in (1.2, 1.4)$
- ▶ R_0 (function of model parameters) suggests initial population drop of 16% to 28% (*compatibility constraint*).
- ▶ Next steps:
 - ▶ Propose $F(R_+, R_-)$, apply to reports from initial outbreak in New York.
 - ▶ Simulate outcomes for initial outbreak in New York from our SEI model.
 - ▶ Calibrate parameters in $F(R_+, R_-)$ to yield population drop (function of R_0) within compatibility constraint.

Modeling surveillance

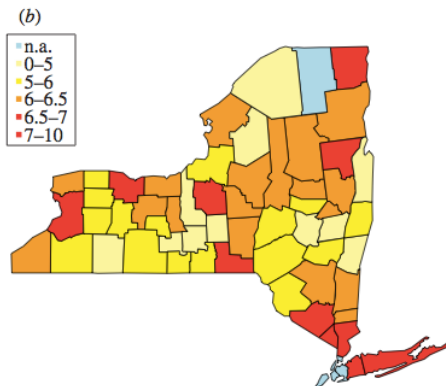
- ▶ Gerardo-Giorda et al. (2013) consider two $F(R_+, R_-)$ surveillance functions.
- ▶ Constant surveillance (function of R_+ and pop'n density h):

$$\begin{aligned}F_{\text{const}}(R_+) &= (1/\gamma)R_+ \\ \gamma &= (1 + K/h)^{-1}\end{aligned}$$

h = population density, $K \uparrow$ reporting rate per density $\downarrow$.

- ▶ Map local K values for each county.

K map (small values good, red = bad)



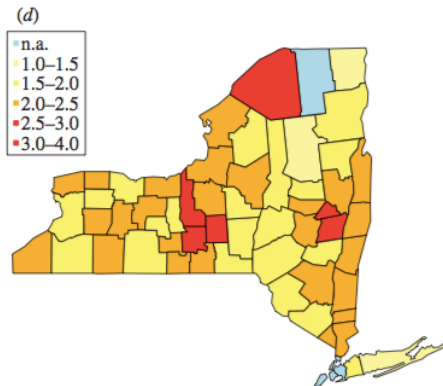
Modeling surveillance (dynamic)

- ▶ Dynamic surveillance (diminishing returns):

$$F_{\text{dyn}}(R_+, R_-) = \left(\frac{N}{R_+ + R_-} \right)^{1/\theta} R_+$$

- ▶ Large θ : high level of surveillance in the area.
- ▶ Small θ : risk that an outbreak could go undetected in this area.
- ▶ Find θ consistent with local R_+ and R_- and meeting compatibility constraint.
- ▶ Map local θ values for each county.

θ map (large values good, red = good)



Surveillance conclusions

- ▶ Good surveillance in some areas under both models, e.g., Tompkins County (Cornell), Albany, but others as well.
- ▶ Poor surveillance in some high population areas, e.g., Onondaga County (Syracuse).

Overall Conclusions

- ▶ Much to be done to link mathematical models to statistical ideas.
- ▶ Disease ecology offers great setting for exploration.
- ▶ Models of transmission, interaction, observation.
- ▶ Mathematical models can inform statistics, statistics can inform models.
- ▶ Room to move past “ad-hockery”.
- ▶ Linking landscape features in a more meaningful (inferential) and spatial way.

Closing

“... Nature’s dice are always loaded, ... in her heaps and rubbish are concealed sure and useful results.”

Ralph Waldo Emerson, *Nature*.

References

- ▶ Smith et al. (2002) Predicting the spatial dynamics of rabies epidemics on heterogeneous landscapes. *PNAS* **99**, 3668-3672.
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- ▶ Gerado-Giorda et al. (2013) Structuring targeted surveillance for monitoring disease emergence by mapping observational data onto ecological process. *J R Soc Interface* 10: 20130418.
- ▶ Wheeler and Waller (2008) Mountains, valleys, and rivers: The transmission of raccoon rabies over a heterogeneous landscape. *JABES* **13**, 388-406.